

Quantum Mekaniği I

Bölüm 6 Tekrar ★ önemli konular

$$i\hbar \frac{d\Psi}{dt} = -\frac{\hbar^2}{2m} \frac{d^2\Psi}{dx^2} = E\Psi \Rightarrow H\Psi = i\hbar \frac{d\Psi}{dt} \text{ zamana bağlı schr}$$

$$i\hbar \frac{d[U(x)T(t)]}{dt} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} U(x)T(t) = E U(x)T(t)$$

$$\Rightarrow i\hbar \frac{dU(x)T(t)}{dt} = E U(x)T(t) \quad \text{1/} U(x)T(t)$$

$$\Rightarrow \frac{i\hbar}{T(t)} \frac{dT(t)}{dt} = E \Rightarrow \frac{dT(t)}{T(t)} = E dt / i\hbar \Rightarrow \ln T = \frac{E}{i\hbar} t + \ln C \Rightarrow \frac{\ln T}{\ln C} = \frac{E}{i\hbar} t \Rightarrow \frac{\ln T}{\ln C} = e^{\frac{E}{i\hbar} t} \Rightarrow T = C e^{\frac{E}{i\hbar} t} \Rightarrow T = C e^{-iEt/\hbar}$$

$$|\Psi(x)|^2 = \int_{-\infty}^{+\infty} dx \Psi^* \Psi = 1 \text{ Normalde}$$

$$\Rightarrow \Psi(x) = \sum_E C_E U_E(x) + \int dE C_E^* U_E \Rightarrow$$

$$\Rightarrow \int dx U_n^*(x) U_{n'}(x) = \sum_{\alpha\beta} x' \cdot x'' = \text{skalar} = \delta_{12} \text{ Kronecker Delta}$$

Kesikli

$$x' \cdot x' = \text{skalar} \quad 1=2=1$$

$$x' \cdot x'' = 0 \Rightarrow x \cdot y = 0 = x \hat{x} \cdot y \hat{y} = x y \hat{x} \hat{y} \quad \hat{x} \cdot \hat{y} = 0$$

$$\int dp U_p^*(x) U_{p'}(x) = \delta(p-p') \text{ sürüklü}$$

$$\int dx \Psi_p^*(x) \Psi_{p'}(x) = \frac{1}{2\pi\hbar} \int e^{i(p-p')x/\hbar} = \delta(p-p') \text{ (Bölm 2. sayfa)}$$

Aşım Post ve Vektör Analizi

$U_\alpha(x)$ ortonormal tam küme $\Rightarrow \Psi$ için

$$\Psi(x) = \sum_\alpha C_\alpha U_\alpha(x) \quad C_\alpha \text{ ları nasıl buluyoruz?}$$

$$\Psi(x) = C_\alpha \sum U_\alpha$$

★ taratı $U_\alpha^*(x)$ ile çarp

$$\Psi^*(x) = C_\alpha^* \sum U_\alpha^*$$

$$\int U_\alpha^*(x) \Psi(x) dx = \sum_\alpha C_\alpha \int dx \underbrace{U_\alpha^*(x) \cdot U_\alpha(x)}_{\delta_{\alpha\alpha}}$$

$$C_\alpha = \int U_\alpha^* \Psi(x) dx \quad \text{B } C_\alpha \text{ bulunma olasılığı.}$$

$$\int dx \Psi^* \Psi = 1 \Rightarrow \int dx (C_\alpha^* \sum U_\alpha^*) (C_\alpha \sum U_\alpha) = \sum_\alpha C_\alpha^* C_\alpha \int dx \underbrace{U_\alpha^* U_\alpha}_{\delta_{\alpha\alpha}}$$

$$\Rightarrow \sum_\alpha |C_\alpha|^2 = 1$$

$$\sum |C_\alpha|^2 = \sum C_\alpha^* C_\alpha = \int dx U_\alpha(x) \Psi^*(x) \int dy U_\alpha^*(y) \Psi(y) dy \Rightarrow$$

$$\Rightarrow \sum \int dx dy U_\alpha(x) U_\alpha^*(y) \Psi^*(x) \Psi(y)$$

$$\Rightarrow \sum U_\alpha(x) U_\alpha^*(y) \int \Psi^*(x) \Psi(y) dx dy \Rightarrow \sum U_\alpha(x) U_\alpha^*(y) = \delta(x-y) \text{ Tamlik Bağıntısı}$$

Operatörler ve Gözlenilme

2 kuantum operatör
lineerlik aksiyomları

$$A(\alpha_1 \psi_1 + \alpha_2 \psi_2) = \alpha_1 A\psi_1 + \alpha_2 A\psi_2 \quad \checkmark \text{ lineer op}$$

Hermitseel op $\langle A \rangle = \langle A^* \rangle$

$$\Rightarrow \langle A \rangle = \int dx \psi^*(x) \{ A \psi(x) \}$$

$$\begin{aligned} \langle A^* \rangle &= \left\{ \int dx \psi^*(x) \{ A \psi(x) \} \right\}^* \Rightarrow \int dx \overbrace{\psi^* A \psi}^{\psi^* A^* \psi} \\ &\Rightarrow \int dx \psi^* A^* \psi = \int dx A \psi^* \psi \\ A &= A^* \Rightarrow \text{Hermitseel op} \end{aligned}$$

$$\langle A \rangle = \int dx \phi^*(x) A \psi(x)$$

$$\begin{aligned} \langle A^* \rangle &\Rightarrow \left\{ \int dx \phi^*(x) A \psi \right\}^* = \int dx \overbrace{\phi^* A \psi}^{\phi^* A^* \psi} \Rightarrow \int dx \psi^* A^* \phi \quad A^* = A \Rightarrow \\ &\Rightarrow \int dx \psi^* A \phi \end{aligned}$$

Dirac Notasyonu

$$\int dx \phi^*(x) \psi(x) =: \langle \phi | \psi \rangle \text{ eğer ki alırsak } \left\{ \int dx \phi^* \psi \right\}^* = \langle \phi | \psi \rangle^* = \langle \psi | \phi \rangle$$

$$\langle \phi | \psi \rangle \text{ eğer ki } = \langle \psi | \phi \rangle$$

aynı $\downarrow$

$$\Rightarrow \int dx \phi^* \psi = 1. \text{ durum}$$

$$1. \text{ durum eğer ki al} \Rightarrow \int dx (\phi^*)^* \psi^* = \int dx \phi \psi^* = \int dx \psi^* \phi = \langle \psi | \phi \rangle$$

$$\langle \phi | \psi \rangle \text{ eğer ki } = \langle \psi | \phi \rangle$$

$$\int dx \phi^* A \psi \text{ eğer ki } = \int dx \psi^* A^* \phi$$

$\downarrow$ (ii)

$$(i) \Rightarrow \int dx \phi^* A \psi = \langle \phi | A \psi \rangle \quad (ii) \int dx \psi^* A^* \phi = \langle \psi | A^* \phi \rangle$$

$$\int dx \phi^* i\alpha \psi = i\alpha \int dx \phi^* \psi \quad \alpha = \text{reel} \quad \alpha^* = \alpha \Rightarrow \langle \phi | B \psi \rangle$$

$$\left(\int dx \phi^* i\alpha \psi \right)^* = \int dx \phi (-i\alpha) \psi^* = \int dx (-i\alpha) \psi^* \phi = +i\alpha \int dx \psi^* \phi = B^* \langle \phi | \psi \rangle$$

$$\int dx i\alpha \phi^* \psi = \langle B \phi | \psi \rangle$$

$$i\alpha = B \quad \alpha^* = \alpha \quad \alpha \text{ reel}$$

$$\left(\int dx i\alpha \phi^* \psi \right)^* = \int dx (-i\alpha) \phi \psi^* = \underbrace{-i\alpha}_{B^*} \int dx \psi^* \phi = B \langle \psi | \phi \rangle$$

Genel kuralı = $\langle \phi | A \alpha \psi \rangle = \langle \alpha^* \phi | A \psi \rangle = \alpha \langle \phi | A \psi \rangle = \alpha \langle A^* \phi | \psi \rangle$

Bu

Dejenerallik ve es Zamanlı Gözlenebilirler

$$A|a\rangle = a|a\rangle \quad A|b\rangle = b|b\rangle$$

$$B|a\rangle = b|a\rangle$$

$$AB|a\rangle = A b|a\rangle = b A|a\rangle = b a|a\rangle$$

$$BA|a\rangle = B a|a\rangle = a B|a\rangle = a b|a\rangle$$

$$AB - BA|a\rangle = 0 \Rightarrow AB - BA = 0 = [A, B] \quad \text{Komütasyon}$$

2 operatör sıradışıysa ortak özfonklara sahip

Belirsizlik Bağıntıları

$$(\Delta A)^2 = \langle (A - \langle A \rangle)^2 \rangle \Rightarrow \langle A^2 - A\langle A \rangle + \langle A \rangle^2 - \langle A \rangle A \rangle$$

$$\Rightarrow \langle A^2 + \langle A \rangle^2 - 2A\langle A \rangle \rangle$$

$$\Rightarrow \langle A^2 \rangle + \langle \langle A \rangle^2 \rangle - \langle 2A\langle A \rangle \rangle$$

$$\langle A^2 \rangle + \langle A^2 \rangle - 2\langle A \rangle^2$$

$$2\langle A^2 \rangle - 2\langle A \rangle^2$$

$$(\Delta A)^2 = \langle A^2 \rangle - \langle A \rangle^2$$

Zaman Bağımlılığı ve Klasik Limit

$$i\hbar \frac{d\Psi}{dt} = H\Psi \Rightarrow \Psi = e^{-iE\Psi/\hbar} U_E(x)$$

$$\langle A \rangle_t = \int dx \Psi^* A \Psi \Rightarrow \frac{d\langle A \rangle}{dt} = \frac{d}{dt} \left(\int dx \Psi^* A \Psi \right) \Rightarrow \frac{d\langle A \rangle}{dt} = \int dx \frac{d\Psi^*}{dt} A \Psi + \int dx \Psi^* \frac{dA}{dt} \Psi + \int dx \Psi^* A \frac{d\Psi}{dt}$$

$$i\hbar \frac{d\Psi}{dt} = H\Psi \Rightarrow \frac{d\Psi}{dt} = \frac{1}{i\hbar} H\Psi \Rightarrow \frac{d\Psi^*}{dt} = \frac{1}{i\hbar} (H\Psi)^* = -\frac{1}{i\hbar} \Psi^* H^\dagger = -\frac{1}{i\hbar} \Psi^* H = \frac{d\Psi^*}{dt}$$

$$\Rightarrow -\frac{1}{i\hbar} \int dx \Psi^* H A \Psi + \langle \frac{dA}{dt} \rangle + \frac{1}{i\hbar} \int dx \Psi^* A H \Psi$$

$$\Rightarrow \langle \frac{dA}{dt} \rangle - \frac{1}{i\hbar} \int dx \Psi^* \{HA - AH\} \Psi = \langle \frac{dA}{dt} \rangle + \frac{1}{i\hbar} \langle [A, H] \rangle$$

$$\Rightarrow A=x \Rightarrow \langle \frac{d}{dt} x \rangle + \frac{1}{i\hbar} \langle [x, H] \rangle = \langle \frac{d}{dt} x \rangle + \frac{1}{i\hbar} \langle [x, \frac{p^2}{2m} + V(x)] \rangle = \frac{d\langle x \rangle}{dt}$$

$$\Rightarrow \frac{1}{i\hbar} \langle [x, \frac{p^2}{2m}] \rangle + \frac{1}{i\hbar} \langle [x, V(x)] \rangle =$$

$$\Rightarrow \frac{1}{i\hbar} \frac{1}{2m} [x, p^2] = \frac{1}{i\hbar} \frac{1}{2m} \{ p [x, p] + [x, p] p \} = \frac{1}{i\hbar} \frac{1}{2m} p^2 \hbar = \frac{p}{m}$$

$$\frac{d\langle x \rangle}{dt} = \frac{1}{m} \langle p \rangle$$

İşte bu yüzden
A = p y
ve p = m v